🐍 Python Tuple — Complete Lesson (Beginner to Intermediate)

I'll teach this like a **Python instructor** — clear, structured, and practical.

1. What is a Tuple?

A **Tuple** is a **collection of items** that is:

- Ordered ✅
- Immutable (cannot be changed) ✅
- Allows duplicate values ✅

Example

```python
numbers = (1, 2, 3, 4, 5)
print(numbers)
```

Output:

```
(1, 2, 3, 4, 5)
```

2. Why Use Tuple Instead of List?

Feature	List	Tuple
Mutable	Yes	No
Faster	No	Yes
Syntax	[]	()
Use Case	Changing data	Fixed data

Example:

```python
list1 = [1,2,3]
tuple1 = (1,2,3)
```

3. Creating Tuple

Method 1: Using Parentheses

```python
t = (10, 20, 30)
print(t)
```

Method 2: Without Parentheses

```python
t = 10, 20, 30
print(t)
```

Output:

```
(10, 20, 30)
```

4. Tuple with Different Data Types

```python
t = (1, "Python", 3.14, True)
print(t)
```

Output:

```
(1, 'Python', 3.14, True)
```

5. Single Element Tuple (Important)

Wrong:

```python
t = (5)
print(type(t))
```

Output:

```
<class 'int'>
```

Correct:

```python
t = (5,)
print(type(t))
```

Output:

```
<class 'tuple'>
```

⚠ **Comma is required**

6. Access Tuple Elements

Indexing works like list.

```python
t = (10, 20, 30, 40)

print(t[0])
print(t[2])
```

Output:

```
10
30
```

7. Negative Indexing

```python
t = (10, 20, 30, 40)

print(t[-1])
print(t[-2])
```

Output:

```
40
30
```

8. Tuple Slicing

```python
t = (10, 20, 30, 40, 50)

print(t[1:4])
```

Output:

```
(20, 30, 40)
```

9. Tuple is Immutable (Cannot Change)

Wrong:

```python
t = (10, 20, 30)
t[0] = 100
```

Error:

```
TypeError: 'tuple' object does not support item assignment
```

10. Loop Through Tuple

Using for loop

```python
t = (10, 20, 30)

for i in t:
    print(i)
```

Output:

```
10
20
30
```

11. Check if Item Exists

```python
t = (10, 20, 30)

print(20 in t)
```

Output:

```
True
```

12. Tuple Length

```python
t = (1, 2, 3, 4)

print(len(t))
```

Output:

```
4
```

13. Tuple Methods (Only 2 Methods)

Because tuple is immutable, it has only **two methods**

1. count()

Counts occurrences

```python
t = (1,2,3,2,2,4)

print(t.count(2))
```

Output:

```
3
```

2. index()

Finds position

```python
t = (10, 20, 30)

print(t.index(20))
```

Output:

```
1
```

14. Tuple Packing

```python
t = 10, 20, 30
print(t)
```

Output:

```
(10, 20, 30)
```

15. Tuple Unpacking

```python
t = (10, 20, 30)

a, b, c = t

print(a)
print(b)
print(c)
```

Output:

```
10
20
30
```

16. Nested Tuple

Tuple inside tuple

```python
t = (1, 2, (3,4,5))

print(t[2])
print(t[2][1])
```

Output:

```
(3, 4, 5)
4
```

17. Convert Tuple to List

```python
t = (1,2,3)

l = list(t)

print(l)
```

Output:

```
[1, 2, 3]
```

18. Convert List to Tuple

```python
l = [1,2,3]

t = tuple(l)

print(t)
```

Output:

```
(1, 2, 3)
```

19. Concatenate Tuple

```python
t1 = (1,2)
t2 = (3,4)

t3 = t1 + t2

print(t3)
```

Output:

```
(1, 2, 3, 4)
```

20. Repeat Tuple

```python
t = (1,2)

print(t * 3)
```

Output:

```
(1, 2, 1, 2, 1, 2)
```

21. Tuple vs List (When to Use Tuple)

Use Tuple When:

✅ Data should not change
✅ Faster performance needed
✅ Fixed data (coordinates, days, months)

Example:

```python
days = ("Sun", "Mon", "Tue", "Wed", "Thu", "Fri", "Sat")
```

Summary

Tuple is:

✅ Ordered
✅ Immutable
✅ Faster than list
✅ Allows duplicates

🧠 Homework

Easy Questions

1. Create a tuple of 5 numbers and print it
2. Access the 3rd element of tuple
3. Find length of tuple

Medium Questions

4. Create tuple:

```
(10,20,30,20,40,20)
```

Count how many times **20** appears.

5. Create nested tuple:

```
(1,2,(3,4,5))
```

Print **4**

Hard Questions

6. Tuple unpacking

```python
t = (100,200,300)
```

Assign values to variables and print them

7. Convert this list to tuple

```
[10,20,30,40]
```

Bonus Challenge (Very Good Practice)

Create tuple of 5 names and print them using loop

Example:

```
("Ram","Hari","Sita","Gita","Shyam")
```

Next lesson (Recommended order):

➡ Dictionary
➡ Set
➡ Functions

If you want, I can also give:

- YouTube script 🎥
- Notes PDF 📄
- Practice questions 🔥